MIL-S-50390(MU)
25 October 1971
SUPERSEDING
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28 May 1965

MILITARY SPECIFICATION

SWITCH AND CONNECTOR ASSEMBLY

1. SCOPE

1.1 This specification is for one type of laminated cable, plastic switch and connector assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-10025	- Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification For
MIL-S-45743	- Soldering Electrical Connections for Guided and Ballistic Missiles, with Electrically-Heated Soldering Irons and Resistance Soldering Apparatus Procedure For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuzes and Fuze Component Environmental and Performance Test For
MIL-STD-1167	- Ammunition Data Card

FSC 1390

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2.1 (Cont'd)

DRAWING

Munitions Command
F10523224 - Switch and Connector Assembly

PUBLICATION

IEL-10523224 - Inspection Equipment List for Switch and Connector Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, reference standards and specifications.

3.1.1 Plastic Materials.

3.1.1.1 Color.-Unless otherwise specified in the contract or order, the plastic material(s) shall be transparent (no coloring matter of any sort to be added).

3.1.1.2 Thermal shock.-The plastic rotors and covers shall withstand thermal shock without fracturing, crazing or deforming when tested in accordance with MIL-STD-331, Test No. 113.

3.2 Resistance.

3.2.1 Insulation resistance.-With all switches closed (monitoring circuit open), the insulation resistance between any 2 circuits (including the function circuits, the monitoring circuit and the line ground circuit) and any circuit to ground shall be 100 megohms, minimum, when measured at 500 ± 50 volts, direct current (dc). (See 4.4.5.2).

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3.2.2 Insulation resistance.-With all switches open, #1 and #2 open, #1 and #3 open, and #2 and #3 open, the insulation resistance between the function circuits shall be 100 megohms, minimum (min), when measured at 500 ± 50 volts, dc. (See 4.4.5.3).

3.2.3 Insulation resistance (Simulated Timer - see IEL).-With all switches open (monitoring circuit closed) the insulation resistance between any 2 circuits (excluding the monitoring circuit and the line ground circuit) and any circuit to ground shall be 100 megohms, min., measured at 500 ± 50 volts, dc. (See 4.4.6 and 4.4.6.1).

3.2.4 Circuit resistance (measured at 100 + 5 milliamperes. See 4.4.5.4).

3.2.4.1 Continuity (All Switches Open).-The resistance between terminals as listed in paragraph 4.4.5.1 shall be 0.10 ohm, max.

3.2.4.2 Monitoring circuit (With All Switches Open).-The circuit resistance of the monitoring circuit shall be 0.7 ohm, max. (See 4.4.5.4.1).

3.2.4.3 Line ground circuit.-The circuit resistance of the line ground circuit shall be 0.01 ohm, max. (See 4.4.5.4.2).

3.2.4.4 Function circuits (With 3 Switches and All Combinations of 2 of 3 Switches Closed).-The circuit resistance of each of the function circuits shall be 0.22 ohm, max. (See 4.4.5.4.3).

3.2.5 Function (Simulated Timer).-The switch and connector assembly when assembled into a simulated timer shall function within the temperature range of minus 40 degrees to plus 125 degrees Fahrenheit (F.) and shall demonstrate the required electrical properties. The rotors shall rotate and shall complete all circuits as required while rotating within the range of 0 to 12,000 rpm after being subjected to linear acceleration of 10,500 g, max. (See 4.4.7).

3.3 Pin location.-The pins in the male connectors shall be in alignment and enter the female socket of a pin location gage without bending or distorting the pins when a pressure of 10 pounds, max. is applied.

3.4 Ballistic (Timer Assembly).

3.4.1 Function.-The switch and connector assembly shall demonstrate the required electrical properties and function within the temperature range of minus 40 degrees to plus 125 degrees (F.) and within the ballistic range of 2000 to 11,500 rpm with linear acceleration of 10,500 g. max., when fired in accordance with the procedure as outlined in IEL. (See Phase II).

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3.5 Environmental.

3.5.1 Jolt.-The switch and connector assembly when assembled into a simulated timer shall comply with MIL-STD-331, Test No. 101.

3.5.2 Jumble.-The switch and connector assembly when assembled into a simulated timer and mounted within a jumble fixture shall comply with MIL-STD-331, Test No. 102.

3.5.3 Transportation-Vibration.-The switch and connector assembly when assembled into a simulated timer shall comply with MIL-STD-331, Test No. 104 and the requirements of 3.2.3 and 3.2.4 through 3.2.4.3.

3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.6.1 Plastic defects.

3.6.1.1 Covers and rotors.-All components shall be free from fractures, crazing, pin burrs and defects, flaking of gold plate of inserts, foreign inclusions and bubbles with burn marks or at a distance less than the greatest diameter of bubble from an insert, pin or surface of plastic.

3.6.1.2 Cables.-All cables shall be free from blisters, creases, splits, or dents. The conductors shall not be bare.

3.6.1.3 Solvents.-No solvents containing aromatic hydrocarbons, chlorinated hydrocarbons, and alkaline substances shall be used in conjunction with any plastic parts.

3.6.2 Metal defects.-All components shall be free from cracks, splits or inclusions.

3.6.3 Threads.-Threads shall be full and undamaged for the entire minimum length as required on the applicable drawing.

3.6.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item, or which might be detrimental to molding of plastic parts.

3.6.5 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings, and shall apply to all surfaces of the component, regardless of configuration.

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3.6.6 Foreign matter.-No part or assembly shall contain chips, dirt, corrosion or other foreign matter.

3.6.7 Soldering.-The soldering shall be uniform and electrically conductive. The joints shall have a smooth metallic appearance and shall show no evidence of excessive solder (buildup above cover), or foreign matter. The soldering procedure and certification of soldering operators shall be in accordance with MIL-S-45743. A copy of the certification shall be furnished the Government.

3.6.8 Serialization.-A serial number shall be assigned and affixed by the contractor to each assembly as required by the applicable drawing, but in no event shall there be any possibility of duplication or any indication of order or size of production quantity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference is made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each inspection lot of switch assemblies shall be comprised of components each produced by one manufacturer by one process in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line.

4.2 First article sample.-When required, the first article sample shall be as specified in the contract and shall consist of the following:

34 (aggregate) switch and connector assemblies as delineated in Phase I and Phase II.

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, thirty-four (34) switch and connector assemblies shall be forwarded to the Timer contractor or such other facility as specified in the contract.

PHASE I

4.2.2 Simulated timers.-Ten (10) of the 34 switch and connector assemblies shall be assembled into simulated (XM32E1) timers and shall be subjected to the following MIL-STD tests and conditions.

Tests	Quantity	Requirement	Method
Jolt	* 2	3.5.1	
Jumble	* 2	3.5.2	
Vibration	* 2	3.5.3	
Monitoring circuit resistance	*	3.2.4.2	4.4.5.4.1
Insulation resistance	*	3.2.3	4.4.6
Airgun, 10,500 g (-40°F)	2	3.2.5	
Function, 12,000 rpm (-40°F)		3.2.5	
Airgun, 10,500 g (+125°F)	2	3.2.5	
Function, 12,000 rpm (+125°F)		3.2.5	
Push test on pins (15 lbs)	All	See Note 1/	

1/ Note - After completion of all tests (10 switch and connector assemblies), a 15 pound push shall be applied to the (20) pins on each connector plug. The test shall be for information purposes to determine security of the pins after soldering and assembly of the cables (See 4.4.4).

4.2.2.1 Acceptance criteria (Phase I).-Any of the following shall be cause for rejection of the first article sample.

a. One or more units fail to comply with the resistance requirement.

b. One or more switch and connector assemblies fail to function or demonstrate the required electrical properties.

c. Failure of more than one switch to function in the combined function tests.

d. The criteria of the applicable MIL-STD shall apply as determined by disassembly and examination in accordance with the procedure in 4.4.8. Where the accept/reject decision is based on engineering judgement, concurrence by the Technical Agency must be obtained.

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4.2.3 Provisional Acceptance.-After completion of tests and provisional acceptance of the 10 switch and connector assemblies in Phase I, the remaining 24 timers shall be assembled into XM32E1 Sequential Timers (see IEL) and ballistically fired in accordance with the provisions of Phase II.

PHASE II

Condition	Quantity	Requirement Paragraph
Ballistic functioned, ambient	8	3.4.1
Ballistic functioned, -40°F	8	3.4.1
Ballistic functioned, +125°F	8	3.4.1

4.2.3.1 Acceptance criteria.-Failure of any switch and connector assembly to comply with the ballistic functional requirements as indicated by timer failure shall be cause for rejection of the first article sample.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a single defect basis in accordance with MIL-STD-105. For each classification of defects contained herein, the AQL given shall be applied to each major and minor defect listed. All nonconforming material shall be rejected. The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.3.2.1 Rivets, Small (after plating) Drawings B10522765, B10522766 and B10522767 covering a detail of Drawing F10523224.

Defect	Method of Inspection
CRITICAL	
None defined	

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MAJOR	AQL = 0.40%	
101.	Eccentricity of head with body	Gage
102.	Diameter of head, incorrect	Gage
103.	Diameter of body, incorrect	Gage
104.	Thickness of head, incorrect	Gage
105.	Protective finish inadequate or defective	Visual
106.	Surface finish on head improper	Visual

4.3.2.2 Rivets, Large (after plating) Drawings B10522768, B10522769 and B10522770 covering a detail of Drawing F10523224.

CRITICAL

None defined

MAJOR	AQL = 0.40%	
101.	Eccentricity of head with body	Gage
102.	Diameter of head incorrect	Gage
103.	Diameter of body incorrect	Gage
104.	Thickness of head incorrect	Gage
105.	Protective finish inadequate or defective	Visual
106.	Surface finish on head, improper	Visual

4.3.2.3 Cups, Contact, Design No. 2, Drawing C10523126 covering a detail of Drawing F10523224.

CRITICAL

None defined

MAJOR	AQL = 0.40%	
101.	Outside diameter (after plating) incorrect	Gage
102.	Inside diameter (after plating) incorrect	Gage
103.	Height (after plating) incorrect	Gage
104.	External profile incorrect	Gage
105.	Protective finish inadequate or defective	Visual
106.	Surface finish improper	Visual
107.	Burr	Visual
108.	Foreign matter	Visual

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4.3.2.4 Pin, Guide, Drawing C10523288 covering a detail of Drawing F10523224.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR	AQL = 0.40%	
101.	Pitch diameter of thread, below min.	Gage
102.	Major diameter of thread, below min.	Gage
103.	Diameter of small end incorrect	Gage
104.	Diameter of large end incorrect	Gage
105.	Total length incorrect	Gage
106.	Length of small end incorrect	Gage
107.	Length of thread incorrect	Gage
108.	Diameter of shoulder incorrect	Gage
109.	Thickness of shoulder incorrect	Gage
110.	Varnish missing or inadequate	Visual
111.	Metal defective	Visual
112.	Thread damaged or not full	Visual
113.	Surface finish, improper	Visual
114.	Burr	Visual

4.3.2.5 Pin (after plating) Drawing B10523289 covering a detail of Drawing F10523224.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR	AQL = 0.40%	
101.	Diameter of male prong incorrect	Gage
102.	Diameter of cavity incorrect	Gage
103.	Diameter of rivet end incorrect	Gage
104.	Diameter of body incorrect	Gage
105.	Diameter of shoulder incorrect	Gage
106.	Length of rivet end incorrect	Gage
107.	Total length incorrect	Gage
108.	Length to flange, rivet end incorrect	Gage
109.	Part missing	Visual
110.	Metal defective	Visual
111.	Protective finish inadequate or defective	Visual
112.	Foreign matter	Visual

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4.3.2.6 Terminal Connector, Drawing C10522772 covering a detail of Drawing F10523224.

Defect		Method of Inspection
CRITICAL		
None defined.		
MAJOR	AQL = 0.40%	
101.	Diameter of pin, both ends (Drawing C10522774) incorrect	Gage
102.	Location of pins incorrect	Gage
103.	Location of inserts incorrect	Gage
104.	Thickness of body incorrect	Gage
105.	Thickness of flange incorrect	Gage
106.	Protrusion of base of pin incorrect	Gage
107.	Plastic defective	Visual
108.	Insert missing or above flush	Visual
109.	Pin scratched or damaged	Visual
110.	Threads damaged or not full	Visual
111.	Radii improper	Visual
112.	Surface finish, improper	Visual
113.	Pin numerals illegible	Visual

4.3.2.7 Spring, Torsion, Drawing C10523136 covering a detail of Drawing F10523224.

CRITICAL		
None defined		
MAJOR	AQL = 0.40%	
101.	Inside diameter incorrect	Gage
102.	Width incorrect	Gage
103.	Thickness	Gage
104.	Location of legs incorrect	Gage
105.	Metal defective	Visual
106.	Burr	Visual

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4.3.2.8 Spring, Contact, Drawing C10523143 covering a detail of Drawing F10523224.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Load at specified height, incorrect	Gage
102.	Solid height, over max.	Gage
103.	Outside diameter at free length incorrect	Gage
104.	Free length incorrect	Gage
105.	Ends incorrect	Visual
106.	Burr	Visual

4.3.2.9 Cable, Ground, Drawing D10522759 covering a detail of Drawing F10523224.

CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Diameter or width of conductor terminal below min.	Gage
102.	Conductor width below min.	Gage
103.	Conductor spacing edge to edge below min.	Gage
104.	Edge of mylar to edge of conductor below min.	Gage
105.	Location of 6 small holes incorrect	Gage
106.	Plastic not removed from 4 terminals as indicated	Visual
107.	Material damaged (creases, splits, dents or conductor exposed)	Visual

4.3.2.10 Rotor, Large, Drawing D10523253 covering a detail of Drawing F10523224.

CRITICAL		
1.	Radial location of rotor pin incorrect	Gage
2.	Angular location of rotor pin incorrect	Gage
3.	Fracture, crazing, pin burr, insert defect (gold flaking) bubbles, or foreign inclusions	Visual (Figures 1 to 14)

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4.3.2.10 (Cont'd)

	Defect	Method of Inspection
MAJOR	AQL = 0.40%	
101.	Perpendicularity of seating surface with center hole incorrect	Gage
102.	Parallelism of both sides of rotor segment with seating surface incorrect	Gage
103.	Diameter of center hole incorrect	Gage
104.	Diameter of seating surface, external, incorrect	Gage
105.	Diameter of seating surface, internal, incorrect	Gage
106.	Diameter of hub incorrect	Gage
107.	Location of holes incorrect	Gage
108.	Thickness of seating surface incorrect	Gage
109.	Thickness through lub incorrect	Gage
110.	Thickness of rotor segment incorrect	Gage
111.	Distance from seating surface to rotor segment incorrect	Gage
112.	Insert protrudes above max.	Gage
113.	Insert(s) not free of plastic	Visual
114.	Surface finish on seating surfaces, hub and face of rotor pin, improper	Visual
115.	Location of drill point incorrect	Gage
116.	Radii missing or improper, except as otherwise classified	Visual

4.3.2.11 Rotor, Small, Drawing D10523254 covering a detail of Drawing F10523224.

CRITICAL		
1.	Radial location of rotor pin incorrect	Gage
2.	Angular location of rotor pin incorrect	Gage
3.	Fracture, crazing, pin burr, insert defect (gold flaking) bubbles or foreign inclusions	Visual (Figures 1 to 14)
MAJOR	AQL = 0.40%	
101.	Perpendicularity of seating surface with center hole incorrect	Gage
102.	Parallelism of both sides of rotor segment with seating surface incorrect	Gage
103.	Diameter of center hole incorrect	Gage

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4.3.2.11 (Cont'd)

	Defect	Method of Inspection
104.	Diameter of seating surface, external, incorrect	Gage
105.	Diameter of seating surface, internal, incorrect	Gage
106.	Diameter of hub incorrect	Gage
107.	Location of holes incorrect	Gage
108.	Thickness of seating surface incorrect	Gage
109.	Thickness through hub incorrect	Gage
110.	Thickness of rotor segment incorrect	Gage
111.	Distance from seating surface to rotor segment incorrect	Gage
112.	Insert protrudes above max.	Gage
113.	Insert(s) not free of plastic	Visual
114.	Surface finish on seating surfaces, hub, and face of rotor pin, improper	Visual
115.	Location of drill point	Gage
116.	Radii missing or improper, except as otherwise classified	Visual

4.3.2.12 Insulator, Drawing D10523287 covering a detail of Drawing F10523224.

CRITICAL

None defined

MAJOR AQL = 0.40%

101.	Diameter of holes incorrect	Gage
102.	Location of holes incorrect	Gage
103.	Thickness incorrect	Gage
104.	Total length incorrect	Gage
105.	Material defective	Visual
106.	Foreign matter	Visual

4.3.2.13 Plate, Drawing D10523292, covering a detail of Drawing F10523224.

CRITICAL

None defined

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4.3.2.13 (Cont'd)

	Defect	Method of Inspection
MAJOR	AQL = 0.40%	
101.	Pitch diameter of thread above max.	Gage
102.	Minor diameter of thread, above max.	Gage
103.	Diameter of holes incorrect	Gage
104.	Diameter of counterbores incorrect	Gage
105.	Diameter of center hole incorrect	Gage
106.	Location of holes incorrect	Gage
107.	Thickness of plate incorrect	Gage
108.	Total length incorrect	Gage
109.	Total height incorrect	Gage
110.	Plastic defective	Visual
111.	Threads damaged or not full	Visual
112.	Varnish missing or inadequate	Visual

4.3.2.14 Connector, Plug Assemblies, Drawings D10523293 and D10523294, covering a detail of Drawing F10523224.

CRITICAL

None defined

MAJOR	AQL = 0.40%	
101.	Perpendicularity of location pin with base of plate incorrect	Gage
102.	Parallelism of locating pins incorrect	Gage
103.	Part missing	Visual
104.	Improper assembly	Visual
105.	Pin(s) scratched or bent	Visual
106.	Plate cracked	Visual
107.	Protective finish inadequate or defective	Visual
108.	Burr	Visual
109.	Foreign matter	Visual
110.	Marking missing, incorrect or illegible	Visual

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4.3.2.15 Cover, Top, Drawing F10523187 covering a detail of Drawing F10523224.

Defect	Method of Inspection
CRITICAL	
1. Fracture, crazing, burr, bubble or foreign inclusions	Visual (Figures 1 to 14)
MAJOR AQL = 0.40%	
101. Perpendicularity of large internal face with datum A incorrect	Gage
102. Perpendicularity of seating face (small) with datum A incorrect	Gage
103. Eccentricity of large inside diameter with datum A	Gage
104. Eccentricity of seating face (small) with datum A	Gage
105. True position of counterbores 2, 4, 6, 8, 10 with datum C incorrect	Gage
106. True position of counterbores 3, 5, 7, 9, 11, 20, with datum B incorrect	Gage
107. Diameter of holes incorrect	Gage
108. Location of holes incorrect	Gage
109. Depth of seating surface (small) internal, incorrect	Gage
110. Thickness overall incorrect	Gage
111. Outside diameter incorrect	Gage
112. Width of keyways incorrect	Gage
113. Location of keyways incorrect	Gage
114. Radii missing or improper	Visual
115. Surface finish S hole, inside diameter and seating surface improper	Visual

4.3.2.16 Cover, Bottom, Drawing F10523188 covering a detail of Drawing F10523224.

Defect	Method of Inspection
CRITICAL	
1. Fracture, crazing, burr, bubble or foreign inclusions	Visual (Figures 1 to 14)

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4.3.2.16 (Cont'd)

	Defect	Method of Inspection
MAJOR	AQL = 0.40%	
101.	Perpendicularity of large internal face with datum A incorrect	Gage
102.	Perpendicularity of seating face (small) with datum A incorrect	Gage
103.	Eccentricity of large inside diameter with datum A	Gage
104.	Eccentricity of seating face (small) with datum A	Gage
105.	True position of counterbores 2, 4, 6, 8, 10 with datum C incorrect	Gage
106.	True position of counterbores 3, 5, 7, 9, 11, 20, with datum B incorrect	Gage
107.	Diameter of holes incorrect	Gage
108.	Location of holes incorrect	Gage
109.	Depth of seating surface (small) internal, incorrect	Gage
110.	Thickness overall incorrect	Gage
111.	Outside diameter incorrect	Gage
112.	Width of keyways incorrect	Gage
113.	Location of keyways incorrect	Gage
114.	Radii missing or improper	Visual
115.	Surface finish S hole, inside diameter and seating surface improper	Visual

4.3.2.17 Laminated Cables No. 1 and No. 2, Drawings F10523225 and F10523226 covering a detail of Drawing F10523224.

CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Diameter or width of conductor terminal below min.	Gage
102.	Conductor width below min.	Gage
103.	Conductor spacing edge to edge below min.	Gage
104.	Edge of mylar to edge of conductor below min.	Gage
105.	Location of holes in short tail incorrect	Gage
106.	Location of holes in long tail incorrect	Gage
107.	Angle of short tail incorrect	Gage
108.	Angle of long tail incorrect	Gage
109.	Plastic not removed from all terminals as indicated	Visual
110.	Material damaged (blisters, creases, splits, dents or conductor exposed)	Visual

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4.3.2.18 Laminated Cables No. 3, 4 and 5, Drawings F10523227, F10523228 and F10523229 covering a detail of Drawing F10523224.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Diameter or width of conductor terminal below min.	Gage
102.	Conductor width below min.	Gage
103.	Conductor spacing edge to edge below min.	Gage
104.	Edge of mylar to edge of conductor below min.	Gage
105.	Location of holes in short tail incorrect	Gage
106.	Angle of short tail incorrect	Gage
107.	Plastic not removed from all terminals as directed	Visual
108.	Material damaged (blisters, creases, splits, dents or conductor exposed)	Visual

4.3.2.19 Laminated Cable No. 6, Drawing F10533934 covering a detail of Drawing F10523224.

CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Diameter or width of conductor terminal below min.	Gage
102.	Conductor width below min.	Gage
103.	Conductor spacing edge to edge below min.	Gage
104.	Edge of mylar to edge of conductor below min.	Gage
105.	Location of holes in long tail incorrect	Gage
106.	Angle of long tail incorrect	Gage
107.	Plastic not removed from all terminals as directed	Visual
108.	Material damaged (blisters, creases, splits, dents or conductor exposed)	Visual

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4.3.2.20 Cable and Cover Assembly, Drawing F10523231 covering a detail of Drawing F10523224.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Misalignment of holes 12 and 13	Gage
102.	Improper assembly	Visual
103.	Soldering improper, missing or excessive	Visual
104.	Cable(s) damaged (blisters, creases, splits, dents)	Visual
105.	Laminated in interconnecting area	Visual
106.	Foreign matter	Visual

4.3.2.21 Switch and Connector Assembly, Drawing F10523224.

CRITICAL		
1.	Engagement of switch rotors (6) with firing arm shaft, below 0.020 min.	Gage
2.	Bent pin(s)	Visual
3.	Broken pin	Visual
MAJOR	AQL = 0.40%	
101.	Missing part(s)	Visual
102.	Cable cut (nicks, gouges, creases, conductor exposed)	Visual
103.	Plating missing or inadequate	Visual
104.	Soldering inadequate, missing, excessive or charred cable	Visual
105.	Foreign matter	Visual
106.	Improper assembly of plugs	Visual
107.	Plastic parts defective, cracks, splits	Visual

4.3.2.22 Packing and Marking (Level C).-The number of specimens and acceptance criteria for packing and marking shall be in accordance with MIL-P-10025.

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4.3.2.23 Pin location.-Each connector plug assembly shall be gaged for proper alignment of the pins. Failure of any unit to comply with the requirement of 3.3 shall be cause for rejection.

4.3.2.24 Monitoring.-Unscheduled monitoring shall be maintained by the inspector to assure the Government that the following requirements are adhered to:

Soldering techniques
Solvents - cleaning plastic parts
Cap screw torque - Drawing F10523224

4.3.3 Inspection testing.-The following tests shall be performed in accordance with provisions of 4.4.

4.3.3.1 Plastic materials.

4.3.3.1.1 Thermal shock.-Two each of the following: small rotor, large rotor, top cover and bottom cover shall be selected at random from each 100 of each part or less in the lot and subjected to the thermal shock. Failure of any piece to comply with the requirement of 3.1.1.2 shall be cause for rejection of the lot of plastic material.

4.3.3.1.2 Certification.-Each batch of plastic materials shall be certified as meeting the requirements of the applicable plastic specification and MIL-Standard. A copy of the certification shall be furnished the Government.

4.3.3.2 Torsion spring.

4.3.3.2.1 Spring force.-(Nondestructive).-Each spring produced shall be subjected to the spring force test. Failure of any spring to comply with the requirement of Drawing C10523136 shall be rejected.

4.3.3.3 Resistance and continuity.

4.3.3.3.1 Insulation resistance.-(Nondestructive).-Each switch and connector assembly produced shall be subjected to insulation and circuit resistance tests. Failure to comply with the requirements of 3.2 through 3.2.4.4 as applicable, shall be cause for rejection.

4.3.3.4 Production lot.

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4.3.3.4.1 Air gun.-Not applicable.

4.3.3.4.2 Spin function (Ambient - 12,000 rpm) (Nondestructive).-Eight switch and connector assemblies from each lot produced shall be subjected to a spin function test. Failure of any unit to comply with the requirements of 3.2.5 shall be cause for rejection of the lot of switch and connector assemblies. (See 6.1).

4.3.4 Inspection equipment.-Index of Inspection Equipment Lists, IEL-10523224 identifies the inspection equipment required to perform the examination and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with the applicable specification referenced on the drawing.

4.4.2 Plastic materials.

4.4.2.1 Thermal shock.-The thermal shock test shall be in accordance with MIL-STD-331, Test No. 113. A light table with four power (4X) magnification shall be used in performing visual examination of the rotors and covers. The Visual Aids, Figures 1 to 14 shall be used as a basis of comparison for defect determination. (See 6.4).

4.4.3 Torsion spring.

4.4.3.1 Spring force.-The test shall be performed on the spring to determine compliance with the requirement of Drawing C10523136.

4.4.4 Connector and plug assembly.

4.4.4.1 Push test.-After the spin test and electrical checks are completed, a 15 lb push shall be applied to the (10) pins on each connector. The push shall be against the shoulder on each pin (except locating) through the rubber and in one direction (setting back).

4.4.5 Resistance and continuity.-All measurements to be made at the terminal pins only.

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4.4.5.1 Continuity (all switches open) - (0.10 ohm, max.).

A3	to	A4	B1	to	B2	C11	to	C12	C17	to	C18
A5	to	A6	B3	to	B4	C13	to	C14	C19	to	C20
A3	to	C15	B5	to	B6	C15	to	C16			

4.4.5.2 Insulation resistance (with all switches closed).-100 megohms minimum when measured at 500 ± 50 volts, dc, with current limited to 5 milliamperes, max.

A1	to	B1	A2	to	A1	A3	to	A1	A5	to	A1	A5	to	C17
A1	to	B3	A2	to	B1	A3	to	B1	A5	to	A2	A5	to	C19
A1	to	B7	A2	to	B5	A3	to	B3	A5	to	A3			
A1	to	B8	A2	to	B7	A3	to	B5	A5	to	B3			
A1	to	B9	A2	to	B8	A3	to	B7	A5	to	B5			
A1	to	B10	A2	to	B9	A3	to	B8	A5	to	B7			
A1	to	C11	A2	to	B10	A3	to	B9	A5	to	B8			
A1	to	C13	A2	to	C11	A3	to	B10	A5	to	B9			
A1	to	C15	A2	to	C13	A3	to	C11	A5	to	B10			
A1	to	C17	A2	to	C15	A3	to	C13	A5	to	C11			
A1	to	C19	A2	to	C17	A3	to	C17	A5	to	C13			
			A2	to	C19	A3	to	C19	A5	to	C15			
B1	to	B3	B3	to	C15	B7	to	C17	B10	to	B7			
B1	to	B5	B3	to	C17	B7	to	C19	B10	to	B9			
B1	to	B8	B3	to	C19				B10	to	C11			
B1	to	B9				B8	to	B5	B10	to	C13			
B1	to	B10	B5	to	B7	B8	to	B7	B10	to	C15			
B1	to	C11	B5	to	B8	B8	to	B9	B10	to	C17			
B1	to	C13	B5	to	B9	B8	to	B10	B10	to	C19			
B1	to	C15	B5	to	B10	B8	to	C11						
B1	to	C17	B5	to	C11	B8	to	C13	C11	to	C13			
B1	to	C19	B5	to	C13	B8	to	C15	C11	to	C15			
			B5	to	C15	B8	to	C17	C11	to	C19			
B3	to	B5	B5	to	C17	B8	to	C19						
B3	to	B7	B5	to	C19				C13	to	C15			
B3	to	B8				B9	to	C11	C13	to	C17			
B3	to	B9	B7	to	B9	B9	to	C13						
B3	to	B10	B7	to	C11	B9	to	C15	C15	to	C17			
B3	to	C11	B7	to	C13	B9	to	C17	C15	to	C19			
B3	to	C13	B7	to	C15	B9	to	C19						
									C17	to	C19			

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4.4.5.3 Insulation resistance.-One hundred megohms min. when measured at 500 ± 50 volts, dc with current limited to 5 milliamperes, max.

a. All switches open
   A5 to B1
   A2 to B3
   A1 to B5
   C11 to C17
   C13 to C19

b. Switches #1 and #3 open
   Test procedure same as (a) above

c. Switches #1 and #2 open
   Test procedure same as (a) above

d. Switches #2 and #3 open
   Test procedure same as (a) above

4.4.5.4 Circuit resistance - Measured at 100 ± 5 milliamperes.-Prior to testing the open circuit voltage of the output of the test equipment shall be 5.5 ± 0.5 volt, dc.

4.4.5.4.1 Monitoring circuit (Pins B7 to B9).-Resistance shall be 0.7 ohm, max. (all switches open).

4.4.5.4.2 Line ground circuit (Pins A3 to C15).-Resistance shall be 0.01 ohm, max.

4.4.5.4.3 Function circuits.-Resistance shall be 0.22 ohm, max.

a. All switches closed
   A5 to B1
   A2 to B3
   A1 to B5
   C11 to C17
   C13 to C19

b. Switches #1 and #2 closed
   Test procedure same as (a) above

c. Switches #2 and #3 closed
   Test procedure same as (a) above

d. Switches #1 and #3 closed
   Test procedure same as (a) above

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4.4.6 Insulation resistance (Simulated Timer).-(With all switches open) - 100 megohms, min. when measured at 500 ± 50 volts, dc. with current listed to 5 milliamperes, max.- The test combinations shall be as specified in paragraph 4.4.5.2 with B7 to B9 deleted and the following added:

A1	to	B5	C11	to	C17
A2	to	B3	C13	to	C19
A5	to	B1			

A1	to	setting mechanism housing	screw
A2	"	"	"
A3	"	"	"
A5	"	"	"
B1	"	"	"
B3	"	"	"
B5	"	"	"
B7	"	"	"
B8	"	"	"
B9	"	"	"
B10	"	"	"
C11	"	"	"
C13	"	"	"
C15	"	"	"
C17	"	"	"
C19	"	"	"

4.4.6.1 Continuity.- (0.01 ohm, max.)

A3	to	A4	B1	to	B2	C11	to	C12
A5	to	A6	B3	to	B4	C13	to	C14
A3	to	C15	B5	to	B6	C15	to	C16
						C17	to	C18
						C19	to	C20

4.4.7 Function.

4.4.7.1 Linear acceleration (10,500 g, max.).-The switch and connector assembly shall be assembled into a simulated timer and tested in the sequence and at the temperature conditions specified in Phase I. An air gun or similar equipment shall be used to perform the test to determine security of the soldered joints.

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4.4.7.2 Spin (12,000 rpm).-The switch and connector assembly shall be assembled into a simulated timer and tested in the sequence and the temperature conditions specified in Phase I and 4.3.3.4.2, as applicable, and while spinning at 12,000 rpm the rotors shall be released and the circuits shall be recorded.

4.4.8 Environmental.

4.4.8.1 Jolt and Jumble.-The tests shall be performed in accordance with MIL-STD-331, Test No. 101 and Test No. 102, as applicable and as outlined for the initial production sample. After breakdown and examination, the units shall be scrapped.

4.4.8.2 Transportation-Vibration (less shipping container).-The simulated timer shall be secured to the vibrator table and the test shall be performed in accordance with MIL-STD-331, Test No. 104. After breakdown and examination, the units shall be scrapped.

4.4.9 Temperature conditioning.

4.4.9.1 Low temperature (-40° ± 5°F.).

4.4.9.2 High temperature (+125° ± 5°F).

4.4.9.3 Conditioning.-The test chamber shall be at ambient temperature at the start of the test and the chamber (cooled or heated) and maintained at a temperature of minus 40°F. or plus 125°F., as applicable until stabilization of the timer temperature is reached. (See 6.2). The timer shall be saturated at minus 40°F. or plus 125°F., as applicable, for one hour minimum.

4.4.10 First article.

4.4.10.1 Ballistic function.-The twenty-four of the 34 switch and connector assemblies shall be assembled into XM32E1 Sequential Timers utilizing movements and components as specified on IEL 10523224. The units shall be temperature conditioned (see Phase II) and ballistically fired as directed by the contracting officer.

5. PREPARATION FOR DELIVERY

5.1 Level A.-Not applicable.

5.2 Level B.-Not applicable.

5.3 Level C.-Packing and Marking.-Unless otherwise specified, packing and marking shall be in accordance with MIL-P-10025.

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6. NOTES

6.1 Ambient as defined herein is room temperature 70 degrees, (F.) plus or minus 10 degrees.

6.2 Temperature stabilization of timers may be checked by a thermal measuring device in good thermal contact with the largest, centrally located internal mass.

6.3 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6.4 Plastic defects defined. (See Visual Aids, Figures 1 to 14).

a. Black area - Bubble appears to have burn marks.

b. Foreign inclusion (Matter molded in plastic part).

6.5 Arming determined by monitoring circuit continuity test, function circuit continuity test.

6.6 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.7 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2632, Rev. 0 dated 28 May 1965.

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Preparing activity

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